

B.TECH

Regular Theory Examination (Odd Sem-V), 2016-17

Solution of Web Technology Paper (NCS-504)

Q1.

a) How do you call a package?

A **java package** is a group of similar types of classes, interfaces and sub-packages.

Package in java can be categorized in two form, built-in package and user-defined package.

There are many built-in packages such as java, lang, awt, javax, swing, net, io, util, sql etc.

Simple example of java package

```
//save as Simple.java
package mypack;
public class Simple{
public static void main(String args[]){
System.out.println("Welcome to package");
}
}
```

To Compile: javac -d .Simple.java

To Run: java mypack.Simple

b) Mention the usage of private access specifier.

Methods, variables, and constructors that are declared private can only be accessed within the declared class itself.

Private access modifier is the most restrictive access level. Class and interfaces cannot be private.

Variables that are declared private can be accessed outside the class, if public getter methods are present in the class.

Using the private modifier is the main way that an object encapsulates itself and hides data from the outside world.

c) What is HTML?

HTML is the standard markup language for creating Web pages.

- HTML stands for Hyper Text Markup Language
- HTML describes the structure of Web pages using markup
- HTML elements are the building blocks of HTML pages
- HTML elements are represented by tags
- HTML tags label pieces of content such as "heading", "paragraph", "table", and so on

- Browsers do not display the HTML tags, but use them to render the content of the page

d) What is selector? Mention its types.

In CSS, selectors are patterns used to select the element(s) we want to style.

Selector	Example	Example description
<i>.class</i>	<code>.intro</code>	Selects all elements with <code>class="intro"</code>
<i>#id</i>	<code>#firstname</code>	Selects the element with <code>id="firstname"</code>
<i>*</i>	<code>*</code>	Selects all elements
<i>element</i>	<code>p</code>	Selects all <code><p></code> elements
<i>element,element</i>	<code>div, p</code>	Selects all <code><div></code> elements and all <code><p></code> elements
<i>element element</i>	<code>div p</code>	Selects all <code><p></code> elements inside <code><div></code> elements
<i>element>element</i>	<code>div > p</code>	Selects all <code><p></code> elements where the parent is a <code><div></code> element
<i>element+element</i>	<code>div + p</code>	Selects all <code><p></code> elements that are placed immediately after <code><div></code> elements
<i>element1~element2</i>	<code>p ~ ul</code>	Selects every <code></code> element that are preceded by a <code><p></code> element
<i>[attribute]</i>	<code>[target]</code>	Selects all elements with a target attribute

e) How a java script works?

JavaScript is called a Client-side Scripting Language. That means that it is a computer programming language that runs inside an Internet browser (a browser is also known as a Web client because it connects to a Web server to download pages).

Inside a normal Web page we place some JavaScript code. When the browser loads the page, the browser has a built-in interpreter that reads the JavaScript code it finds in the page and runs it.

f) Differentiate Static web pages and Dynamic Web pages.

Static Web Pages

Static Web pages display the exact same information whenever anyone visits it. Static Web pages do not have to be simple plain text. They can feature detailed multimedia design and even videos.

However, every visitor to that page will be greeted by the exact same text, multimedia design or video

every time he visits the page until you alter that page's source code.

Dynamic Web Pages

Dynamic Web pages are capable of producing different content for different visitors from the same source code file. The website can display different content based on what operating system or browser the visitor is using, whether she is using a PC or a mobile device, or even the source that referred the visitor. A dynamic Web page is not necessarily better than a static Web page. The two simply serve different purposes.

g) What is Java Server Pages Technology?

JavaServer Pages (JSP) is a server-side programming technology that enables the creation of dynamic, platform-independent method for building Web-based applications. JSP have access to the entire family of Java APIs, including the JDBC API to access enterprise databases.

h) Compare Servlet with CGI.

Servlet	CGI (Common Gateway Interface)
Servlets are portable	CGI is not portable.
In Servlets each request is handled by lightweight Java Thread	IN CGI each request is handled by heavy weight OS process
In Servlets, Data sharing is possible	In CGI, data sharing is not available.
Servlets can link directly to the Web server	CGI cannot directly link to Web server.
Session tracking and caching of previous computations can be performed	Session tracking and caching of previous computations cannot be performed
Automatic parsing and decoding of HTML form data can be performed.	Automatic parsing and decoding of HTML form data cannot be performed.
Servlets can read and Set HTTP Headers	CGI cannot read and Set HTTP Headers
Servlets is inexpensive than CGI	CGI is more expensive than Servlets

I) What is the usage of CSS?

CSS is used to define styles for documents, including the design, layout and variations in display for different devices and screen sizes. We can place CSS in the <head> of a document with an embedded style sheet, or attach a separate file that defines the styles with an external style sheet. To link an external style sheet to the document, we simply add a link to the style sheet in the <head> of the document.

An external style sheet has many advantages. Keeping the styles separate from your HTML content:

- Helps avoid duplication
- Makes maintenance easier
- Allows you to make a site-wide change in one place

j) Define ODBC.

Open Database Connectivity (ODBC) is a standard application programming interface (API) for accessing database management systems (DBMS). The designers of ODBC aimed to make it independent of database systems and operating systems. An application written using ODBC can be ported to other platforms, both on the client and server side, with few changes to the data access code.

SECTION B

a) Briefly explain the Internet concepts

The **Internet** is the global system of interconnected computer networks that use the Internet protocol suite (TCP/IP) to link devices worldwide. It is a *network of networks* that consists of private, public, academic, business, and government networks of local to global scope, linked by a broad array of electronic, wireless, and optical networking technologies. The Internet carries an extensive range of information resources and services, such as the inter-linked hypertext documents and applications of the World Wide Web (WWW), electronic mail, telephony, and peer-to-peer networks for file sharing.

What is Internet?

The Internet is essentially a global network of computing resources. You can think of the Internet as a physical collection of routers and circuits as a set of shared resources.

Some common definitions given in the past include –

- A network of networks based on the TCP/IP communications protocol.
- A community of people who use and develop those networks.
- A community of people who use and develop those networks.

Internet-Based Services

Some of the basic services available to Internet users are –

- **Email** – A fast, easy, and inexpensive way to communicate with other Internet users around the world.
- **Telnet** – Allows a user to log into a remote computer as though it were a local system.
- **FTP** – Allows a user to transfer virtually every kind of file that can be stored on a computer from one Internet-connected computer to another.

- **UseNet news** – A distributed bulletin board that offers a combination news and discussion service on thousands of topics.
- **World Wide Web (WWW)** – A hypertext interface to Internet information resources.

WWW stands for **World Wide Web**. A technical definition of the World Wide Web is – All the resources and users on the Internet that are using the Hypertext Transfer Protocol (HTTP).

A broader definition comes from the organization that Web inventor Tim Berners-Lee helped found, the World Wide Web Consortium (W3C): The World Wide Web is the universe of network-accessible information, an embodiment of human knowledge.

In simple terms, The World Wide Web is a way of exchanging information between computers on the Internet, tying them together into a vast collection of interactive multimedia resources.

HTTP stands for **H**ypertext **T**ransfer **P**rotocol. This is the protocol being used to transfer hypertext documents that makes the World Wide Web possible.

A standard web address such as Yahoo.com is called a URL and here the prefix **http** indicates its protocol

URL stands for **U**niform **R**esource **L**ocator, and is used to specify addresses on the World Wide Web. A URL is the fundamental network identification for any resource connected to the web (e.g., hypertext pages, images, and sound files).

A URL will have the following format –

protocol://hostname/other_information

The protocol specifies how information is transferred from a link. The protocol used for web resources is HyperText Transfer Protocol (HTTP). Other protocols compatible with most web browsers include FTP, telnet, newsgroups, and Gopher.

The protocol is followed by a colon, two slashes, and then the domain name. The domain name is the computer on which the resource is located.

Links to particular files or subdirectories may be further specified after the domain name. The directory names are separated by single forward slashes.

Each page available on the website is called a *web page* and first page of any website is called *home page* for that site.

Every Website sits on a computer known as a Web server. This server is always connected to the internet. Every Web server that is connected to the Internet is given a unique address made up of a series of four numbers between 0 and 256 separated by periods. For example, 68.178.157.132 or 68.122.35.127.

When you register a Web address, also known as a domain name, such as tutorialspoint.com you have

to specify the IP address of the Web server that will host the site.

We will see different type of Web servers in a separate chapter.

Web Browsers are software installed on your PC. To access the Web you need a web browsers, such as Netscape Navigator, Microsoft Internet Explorer or Mozilla Firefox.

Currently you must be using any sort of Web browser while you are navigating through my site tutorialspoint.com. On the Web, when you navigate through pages of information this is commonly known as *browsing or surfing*.

SMTP stands for **S**imple **M**ail **T**ransfer **P**rotocol Server. This server takes care of delivering emails from one server to another server. When you send an email to an email address, it is delivered to its recipient by a SMTP Server.

ISP stands for **I**nternet **S**ervice **P**rovider. They are the companies who provide you service in terms of internet connection to connect to the internet.

You will buy space on a Web Server from any Internet Service Provider. This space will be used to host your Website.

HTML stands for **H**yper **T**ext **M**arkup **L**anguage. This is the language in which we write web pages for any Website. Even the page you are reading right now is written in HTML.

This is a subset of Standard Generalized Mark-Up Language (SGML) for electronic publishing, the specific standard used for the World Wide Web.

A hyperlink or simply a link is a selectable element in an electronic document that serves as an access point to other electronic resources. Typically, you click the hyperlink to access the linked resource. Familiar hyperlinks include buttons, icons, image maps, and clickable text links.

DNS stands for **D**omain **N**ame **S**ystem. When someone types in your domain name, www.example.com, your browser will ask the Domain Name System to find the IP that hosts your site. When you register your domain name, your IP address should be put in a DNS along with your domain name. Without doing it your domain name will not be functioning properly.

W3C stands for **W**orld **W**ide **W**eb Consortium which is an international consortium of companies involved with the Internet and the Web.

The W3C was founded in 1994 by Tim Berners-Lee, the original architect of the World Wide Web. The organization's purpose is to develop open standards so that the Web evolves in a single direction rather than being splintered among competing factions. The W3C is the chief standards body for HTTP and HTML.

Benefits of the Internet

1. Helps in reducing the costs of communication, as one can very easily be in touch and communicate

with each other with the help of chatting, electronic mail etc.

2. Acts as a very strong source or a tool at the corporate level, as it helps, inter alia, brings business and helps in generating revenue.
3. Communication and also the coordinating ability between the individuals have been enhanced.
4. Customer satisfaction and customer service have been improved at a very large scale.
5. One can very easily go through or access the various data or the information needed by an individual, hence having a very positive effect in speeding up the whole process of the distribution of knowledge.

Disadvantages of the Internet

1. Legal problems.
2. Problems related to security.
3. Technological problems.
4. Lacking behind in standards as a result of following the same traditional internet culture.

b) Compare and contrast java and javascript

The JavaScript programming language, developed by Netscape, Inc., is not part of the Java platform.

JavaScript **does not create applets or stand-alone applications**. In its most common form, JavaScript **resides inside HTML documents**, and can provide levels of **interactivity to web pages** that are not achievable with simple HTML.

Key differences between Java and JavaScript:

- Java is an OOP **programming language** while Java Script is an OOP **scripting language**.
- Java creates applications that **run in a virtual machine** or browser while JavaScript code is **run on a browser only**.
- Java code needs to be **compiled** while JavaScript code are **all in text**.
- They require different plug-ins.

	Java	JavaScript
Designed by	James Gosling and Sun Microsystems	Brendan Eich
Major Implementations	OpenJDK, many others	Netscape
Typing Discipline	Static, strong, safe, nominative, manifest	dynamic, weak, duck
Appeared in	1995	1995; 18 years ago
Influenced by	Ada 83, C++, C#, Eiffel, Generic Java, Mesa, Modula-3, Oberon, Objective-C, UCSD Pascal, Smalltalk	C, Java, Perl, Python, Scheme, Self
Influenced	Ada 2005, BeanShell, C#, Clojure, D, ECMAScript, Groovy, J#, Javascript, PHP, Python, Scala, Seed7, Vala	ActionScript, CoffeeScript, Dart, JScript .NET, Objective-J, QML, TIScript, TypeScript

code	compiled to bytecode	Codes are all in text, does not need to be compiled
Statement terminators	;	Semicolon terminated
Runs on	Virtual machine	web-browser
Stable release	Java Standard Edition 8 (1.8) / March 18, 2014	1.8.5 (March 22, 2011; 2 years ago)
Introduction (from Wikipedia)	Java is a computer programming language that is concurrent, class-based, object-oriented, and specifically designed to have as few implementation dependencies as possible.	JavaScript is an interpreted computer programming language.

c) **Design a HTML form for a railway reservation system**

<HTML>

<BODY>

<FORM>

<H1>RESERVATION FORM</H1>

<P>NAME<INPUT TYPE="TEXT" SIZE="20" MAXLENGTH="99" />

<P>AGE<INPUT TYPE="TEXT" SIZE="12" MAXLENGTH="99" />

<P>SEX

<SELECT NAME="sex" >

<OPTION>MALE</OPTION>

<OPTION>FEMALE</OPTION></SELECT>

<P>ADDRESS<INPUT

<TYPE="TEXT" SIZE="42" MAXLENGTH="99" />

<P>EMAIL<INPUT TYPE="TEXT" SIZE="30" MAXLENGTH="99" />


```

<P>TELEPHONE<INPUT TYPE="TEXT" SIZE="10" MAXLENGTH="99" />

<P>SELECT YOUR BERTH

<SELECT NAME="CHOICES" SIZE="3">

<OPTION>TATKAL</OPTION>

<OPTION>LADIES</OPTION>

<OPTION>GENRAL</OPTION>

</SELECT></P>

<P>CITY<SELECT naME="CITY"><OPTION SELECTED>

DELHI</OPTION><OPTION>MUMBAI</OPTION><OPTION>KOLKATA</OPTION><OPTION
>CHENNAI</OPTION></p><br><br>

<input type="submit" value="click here to submit" />

<input type="reset" value="clear this form" />

</FORM>

</BODY></HTML>

```

d) Explain in detail the Tomcat Server

Apache Tomcat, often referred to as **Tomcat Server**, is an open-source Java Servlet Container developed by the Apache Software Foundation (ASF). Tomcat implements several Java EE specifications including Java Servlet, JavaServer Pages (JSP), Java EL, and WebSocket, and provides a "pure Java" HTTP web server environment in which Java code can run.

Tomcat is developed and maintained by an open community of developers under the auspices of the Apache Software Foundation, released under the Apache License 2.0 license, and is open-source software. **Apache Tomcat**, is an open source web server and servlet container developed by the Apache SoftwareFoundation. Basically, it implements the Java Servlet and the JavaServer Pages (JSP) specifications from Sun Microsystem, and provides a "pure Java" HTTP web server environment for Java code to run in. In the simplest configuration Tomcat runs in a single operating system process. The process runs a Java virtual machine (JVM) and every single HTTP request from a browser to Tomcat is processed in the Tomcat process in a separate thread. Apache Tomcat includes tools for configuration and management, but can also be configured by

editing

XML

configuration

files.

In simple words, Tomcat acts or behaves as a *development* server on your desktop to use for testing when building applications that use JSF 2, servlets/JSP, or other Java-based dynamic Web technologies. Regardless of what deployment server you use, you'll want a standalone server on your desktop to use for development. By far the best way to use Tomcat for development purposes is from inside Eclipse or another IDE.

e) What do you understand by ASP? What are its advantages and disadvantages?

Active Server Pages

Active Server Pages or ASP.NET is a new programming framework from Microsoft for deploying next generation web applications. It is a structure built on the Common Language Runtime and introduces a new model to server-side web development. It is used with IIS (Internet Information Server) which runs on Microsoft servers.

Advantages of ASP

It provides 3-tiers architecture.

Provide easy way to deploy the new applications and maintenance.

Compatible with around 55 languages.

Consistent Programming Model.

Direct Support for Security.

Rapid development to web applications.

Easy to learn and install.

Disadvantages of ASP

Cost Factor

Running ASP programs costs higher as it requires IIS installed on a Windows platform server which is not free. Moreover, the connectivity of the database is very costly in ASP as you have to buy MS-SQL which is a Microsoft product.

Speed

When it comes to speed, ASP is relatively slower than PHP. This is because ASP can't run fast as it is built on COM based architecture which is an overhead for the server.

Platform Compatibility

ASP is not platform compatible like PHP as and is primarily related with Windows platform with ASP-Apache installed on the server.

Additional Cost

Unlike PHP ASP doesn't have any built-in features such as FTP, email from a webpage and encryption mechanisms and hence, some additional components are required which results in additional expenditure as well.

Base Language

ASP has lots of syntax that are based on Visual Basic which most programmers are not conversant with. Moreover, it is again a product of Microsoft. Probably, this is why many coders prefer PHP over ASP.

ASP has always been a prime choice for e-commerce application development. The controversy as to which one is the better between ASP and PHP still continues. However, both have their respective merits and demerits. Either of these programming languages supports a specific platform. Hence, it is really difficult to say that one is better than the other.

- **Interpreted and Loosely-Typed Code**

ASP scripting code is usually written in languages such as JScript or VBScript. The script-execution engine that Active Server Pages relies on interprets code line by line, every time the page is called. In addition, although variables are supported, they are all loosely typed as variants and bound to particular types only when the code is run. Both these factors impede performance, and late binding of types makes it harder to catch errors when you are writing code.

- **Mixes layout (HTML) and logic (scripting code)**

ASP files frequently combine script code with HTML. This results in ASP scripts that are lengthy, difficult to read, and switch frequently between code and HTML. The interspersing of HTML with ASP code is particularly problematic for larger web applications, where content must be kept separate from business logic.

- **Limited Development and Debugging Tools**

Microsoft Visual InterDev, Macromedia Visual UltraDev, and other tools have attempted to increase the productivity of ASP programmers by providing graphical development environments. However, these tools never achieved the ease of use or the level of acceptance achieved by Microsoft Windows application development tools, such as Visual Basic or Microsoft Access. ASP developers still rely heavily or exclusively on Notepad.

Debugging is an unavoidable part of any software development process, and the debugging tools for ASP have been minimal. Most ASP programmers resort to embedding temporary Response.Write statements in their code to trace the progress of its execution.

- **No real state management**

Session state is only maintained if the client browser supports cookies. Session state information can

only be held by using the ASP Session object. And you have to implement additional code if you, for example, want to identify a user.

- **Update files only when server is down**

If your Web application makes use of components, copying new files to your application should only be done when the Web server is stopped. Otherwise it is like pulling the rug from under your application's feet, because the components may be in use (and locked) and must be registered.

- **Obscure Configuration Settings**

The configuration information for an ASP web application (such as session state and server timeouts) is stored in the IIS metabase. Because the metabase is stored in a proprietary format, it can only be modified on the server machine with utilities such as the Internet Service Manager. With limited support for programmatically manipulating or extracting these settings, it is often an arduous task to port an ASP application from one server to another.

f) Explain COM and DCOM in detail

COM: Component object model is an interface standard for software component introduced by Microsoft in 1993. It is used to enable interprocess communication and dynamic object creation in any programming language that supports the technology. The term COM is often used in the software development industry as an umbrella term that encompasses the OLE, OLE Automation, ActiveX, COM+ and DCOM technologies.

COM is a software architecture that allows applications and systems to be built from the components supplied by different software vendors. It is a set of binary and network standards that allows any software to communicate with each other regardless of the hardware, the operating system, and programming language used for development. COM is not a Programming language but a specification that defines how components can communicate with each other.

In a component based system, components interact with each other by calling methods and passing data. COM ensures that there is a standard method of interaction between the components. All the COM objects need to follow these standards when providing the functionality. Standards are as important to component architecture as they are to any system with interchangeable parts. For example, the signal that a television or radio receives is specified in a standards. Without standards, nothing will function together.

Benefits of COM:

Any Programming language that can create structures of pointers and call functions through pointers can create and use components objects. Components can be implemented in a number of programming languages and can be used for client software written in different programming languages. **This is because COM** provides a binary and not source code standard. This is the fundamental benefit of COM over object oriented language.

DOCM: DCOM is a model as COM but is specially designed for distributed applications. While there is a need of using the applications as distributed then DCOM becomes in picture.

Distributed Component Object Model is proprietary Microsoft technology for communication among software components distributed across networked computers. DCOM, which originally was called “Network OLE”, extends Microsoft’s COM, and provides the communications substrate under Microsoft COM+ application sever infrastructure. It has been deprecated in favour of Microsoft .Net.

In terms of the extensions it added to COM, DCOM had to solve the problem of

- 1. Marshalling:** serialize and de-serializing the arguments and returns values of methods calls over the wire.
- 2. Distributed garbage collection:** ensuring that references held by clients of interfaces are released when, for example, the client process crashed, or the network connection was lost.

g) Describe the PHP life cycle and how does PHP score over ASP and JSP?

Life Cycle of PHP

1. We never start any PHP daemon or anything by ourself. When we start Apache, it starts the PHP interpreter along itself
2. PHP is linked to Apache (In general term SAPI i.e. a Server API) using mod_php5.so module
3. PHP as a whole consists of 3 modules (Core PHP, Zend Engine and Extension Layer)
4. Core PHP is the module which handles the requests, file streams, error handling and other such operations
5. Zend Engine(ZE) is the one which converts human readable code into machine understandable tokens/op-codes. Then it executes this generate code into a Virtual Machine.
6. Extensions are a bunch of functions, classes, streams made available to the PHP scripts, which can be used to perform certain tasks. For example, we need mysql extension to connect to MySQL database using PHP.
7. While Zend Engine executes the generated code, the script might require access to a few extensions. Then ZE passes the control to the extension module/layer which transfer back the control to ZE after completion of tasks.

8. Finally Zend Engine returns back the result to PHP Core, which gives that to SAPI layer, and finally which displays it on your browser.

Score PHP :

PHP

PHP uses a blend of interpretation and compilation in order to provide the best mix of performance and flexibility to programmers.

Behind the scenes, PHP compiles your script down to a series of instructions (called opcodes) whenever it is accessed. These instructions are then executed one by one until the script terminates. This is different from conventional compiled languages such as C++ where the code is compiled down to native executable code then that executable is run from then on. Instead, PHP re-compiles your script each time it is requested.

This constant recompilation may seem a waste of processor time, but it is actually not all that bad because you no longer need to worry about hand recompiling your scripts when you make any changes. On the flip side, many scripts take longer to compile than they do to execute!

Furthermore, it provides very quick feedback during development. If you have an error somewhere in your file, PHP will refuse to compile the page until you have fixed the problem, and you are able to step through execution of your code line by line until you find the problem.

The speed hit of regular compilation is nullified entirely by the use of PHP accelerators.

One major advantage to having interpreted code is that all memory used by the script is managed by PHP, and the language automatically cleans up after every script has finished. This means that you do not need to worry about closing database links, freeing memory assigned to images, and so on, because PHP will do it for you. That is not to say you should be lazy and make PHP do all the work – good programmers clean up themselves, and let PHP work as backup in case something is missed.

- ASP only runs on IIS installed on a Windows platform server
- PHP code runs faster than ASP.
- ASP is built on COM based architecture, which is an overhead for the server whereas PHP code runs in its own memory space.
- PHP programs can run on various platforms like Linux, Unix, Windows and Solaris whereas ASP is mainly associated with Windows platforms.
- PHP has inbuilt features like ftp, email from a web page or even encryption mechanisms but in ASP such features are not built in.
- PHP is based on C++ language and the syntax used in PHP is quite similar to C/C++. ASP's syntax is based on the syntax of Visual Basic.
- PHP can connect to various databases, mainly MySQL, whereas ASP mainly uses MS-SQL.
- PHP codes tend to be easier than ASP codes.

PHP and JSP

PHP can be expanded as **Hypertext Preprocessor**. PHP is a scripting language that is extensively used in the area of Web development. It allows the web developers to quickly write dynamically generated pages.

JSP is JavaServer Pages. It is a technological innovation from JAVA that enables Web developers and designers to develop and maintain, dynamic Web pages that are rich in information which in turn, empower systems.

PHP	JSP
It is the most widely used Web development platform in the world. The barrier to entry is really low.	It is used in broader areas and barrier is higher than PHP.
For this need Server as (typically Apache).	For this need server as Apache Tomcat or JBoss or WebSphere, WebLogic etc.
Immediate feedback and greater productivity as coding less and fast development and execution.	As more coding than PHP need little more time than PHP.
PHP hosting is extraordinarily cheap.	JSP hosting is not much costly than PHP.
Statically typed. Now I don't consider this a \"win\" against dynamic typing, they're just different. But the fact is that if you mistype a variable name in Java it'll not throw any error and will generate a new variable.	Statically typed. Now I don't consider this a \"win\" against dynamic typing, they're just different. But the fact is that if you mistype a variable name in Java it'll generate a compiler error.
Less libraries compare to JSP.	There are a huge amount of third party libraries for Java Web development out there.
Less robust compare to JSP.	IMHO the modern Java Web development approach using, say, Spring (or other IoC container) is extremely robust.
Multithreaded and background processing are somewhat kludgy in PHP.	Multithreaded and background processing are far more natural in Java.
PHP does not allow you to define custom tags.	JSP allow you to define custom tags that can be quite powerful.

Here is a list of reasons why JSP is better than PHP:

- Anything you can do with PHP, you can do with JSP; the reverse is not true
- JSP is much more powerful, since it has access to all the Java libraries. PHP only has access to PHP libraries
- JSP is Object-Oriented, so leads to cleaner code that's easier to debug, maintain, and improve. (PHP also allows objects, but the object model is more primitive, and most scripted pages ignore PHP objects and just use normal variables.)
- The equivalent syntax in JSP is just as simple to learn, so you can get up and running just as quickly -- that is, there's no **extra** startup cost in using Java, at least not a significant one
- Java programmers (as opposed to 15-year-old hackers or HTML monkeys) appreciate the importance of a clean language with complex OO data structures and strong typing

- With JSP, if the code inside a page gets too big, or if you want to use it elsewhere, you can cut it out, make it into a Java class, and invoke it from anywhere in your application (even not from a page). With PHP, you're stuck inside the HTML box.
- JSP's concept of state management and persistence is more explicit and powerful than PHP's. With JSP, you can specify whether a variable persists for the page, the request, the session, or the application (or if it's just local to the function). The JSP engine automatically does the right thing with cookies so you have access to the variable on later requests. With PHP, you just have "global" and "not global", you don't have automatic session management, and have to do your state thing manually with cookies or hidden variables.

SECTION-C

Q3. Elucidate the features of Java.

i. Simple

According to Sun, Java language is simple because:

syntax is based on C++ (so easier for programmers to learn it after C++).

removed many confusing and/or rarely-used features e.g., explicit pointers, operator overloading etc.

No need to remove unreferenced objects because there is Automatic Garbage Collection in java.

ii. Object-oriented

Object-oriented means we organize our software as a combination of different types of objects that incorporates both data and behaviour.

Object-oriented programming(OOPs) is a methodology that simplify software development and maintenance by providing some rules.

Basic concepts of OOPs are:

1. Object
2. Class
3. Inheritance
4. Polymorphism
5. Abstraction
6. Encapsulation

iii. Platform Independent

A platform is the hardware or software environment in which a program runs.

There are two types of platforms software-based and hardware-based. Java provides software-based platform.

The Java platform differs from most other platforms in the sense that it is a software-based platform that runs on the top of other hardware-based platforms. It has two components:

1. Runtime Environment
2. API(Application Programming Interface)

Java code can be run on multiple platforms e.g. Windows, Linux, Sun Solaris, Mac/OS etc. Java code

is compiled by the compiler and converted into bytecode. This bytecode is a platform-independent code because it can be run on multiple platforms i.e. Write Once and Run Anywhere(WORA).

iv. Secured

Java is secured because:

- **No explicit pointer**
- **Java Programs run inside virtual machine sandbox**
- **ClassLoader:** adds security by separating the package for the classes of the local file system from those that are imported from network sources.
- **Bytecode Verifier:** checks the code fragments for illegal code that can violate access right to objects.
- **Security Manager:** determines what resources a class can access such as reading and writing to the local disk.

These security are provided by java language. Some security can also be provided by application developer through SSL, JAAS, Cryptography etc.

v. Robust

Robust simply means strong. Java uses strong memory management. There are lack of pointers that avoids security problem. There is automatic garbage collection in java. There is exception handling and type checking mechanism in java. All these points makes java robust.

vi. Architecture-neutral

There is no implementation dependent features e.g. size of primitive types is fixed.

In C programming, int data type occupies 2 bytes of memory for 32-bit architecture and 4 bytes of memory for 64-bit architecture. But in java, it occupies 4 bytes of memory for both 32 and 64 bit architectures.

vii. Portable

We may carry the java bytecode to any platform.

viii. High-performance

Java is faster than traditional interpretation since byte code is "close" to native code still somewhat slower than a compiled language (e.g., C++)

ix. Distributed

We can create distributed applications in java. RMI and EJB are used for creating distributed applications. We may access files by calling the methods from any machine on the internet.

x. Multi-threaded

A thread is like a separate program, executing concurrently. We can write Java programs that deal with many tasks at once by defining multiple threads. The main advantage of multi-threading is that it doesn't occupy memory for each thread. It shares a common memory area. Threads are important for multi-media, Web applications etc.

Q4. What is XML? Discuss the significance of XML. How is XML different from HTML? Explain the process of parsing. How are they useful?

XML stands for eXtensible Markup Language. The syntax rules of XML are very simple and very strict. The rules are very easy to learn, and very easy to use. Because of this, creating software that can read and manipulate XML is very easy to do.

An Example XML Document

XML documents use a self-describing and simple syntax. For example, the following is a complete XML file presenting a note:

```
<?xml version="1.0" encoding="ISO-8859-1"?>
<note>
<to>Tove</to>
<from>Jani</from>
<heading>Reminder</heading>
<body>Don't forget me this weekend!</body>
</note>
```

The first line in the document – the XML declaration – defines the XML version and the character encoding used in the document. In this case the document conforms to the 1.0 specification of XML and uses the ISO-8859-1 (Latin-1/West European) character set.

The next line and the last line describe the root element of the document (like it was saying: “this document is a note”).

Significance of XML-

XML was designed to describe data while HTML was designed for displaying data. If you remember, HTML tags control the way data is presented to browser users, color, font size, spacing, etc. Differently XML aims to deal with the logic meaning of data, or semantics. In the above example, the text wrapped in <from> and </from> is the name of the sender of the note. This enables the fulfillment of the task of finding all the notes written by a specific person. So XML was designed to describe data and to focus on what data is while HTML was designed to display data and to focus on how data looks. Actually XML and HTML can complement each other. For example, we use XML files to store data on a web server machine and when a request arrives, a servlet runs to retrieve data in XML, compose a HTML file, and finally output it to the client. This way you can concentrate on using HTML for data layout and display, and be sure that changes in the underlying data will not require any changes to your HTML.

Besides XML's role of storing data, with XML, data can be exchanged between incompatible systems. In the real world, computer systems and databases contain data in incompatible formats.

One of the most time-consuming challenges for developers has been to exchange data between such systems over the Internet. Converting the data to XML can greatly reduce this complexity and create data that can be read by many different types of applications, since XML data is stored in plain text format, which is software- and hardware-independent. The best description of XML may be this: XML is a cross-platform, software and hardware independent tool for transmitting information. Since the creation of XML, it has been amazing to see how quickly the XML standard has been developed and how quickly a large number of software vendors have adopted the standard. It is strongly believed by the IT community at large that XML will be as important to the future of the Web as HTML has been

to the foundation of the Web and that XML will be the most common tool for all data manipulation and data transmission.

It is a markup language much like HTML, but there are several differences between them:

- HTML includes a collection of predefined tags that you can use right away in editing your HTML files, e.g. and <h1>, but XML tags are not predefined. To use XML, users have to define their own tags for their specific application before using them. For example, to describe a note, <note>, <to>, <from>, <heading>, and <body> are defined in advance and used in a nested fashion to present the following XML file as a note:

```
<note>
<to>Tove</to>
<from>Jani</from>
<heading>Reminder</heading>
<body>Don't forget the party!</body>
</note>
```

With XML, one can define whatever tags needed, which together compose a user-defined markup language similar to HTML, and then use the language to describe data. Specifically XML uses Document Type Definition (DTD) or an XML Schema to define tags. In this sense, XML is viewed as a meta-language since it can be used to define and describe a markup language instead of concrete data directly. That is also why it is called extensible.

- Different from HTML, XML has a syntax that is much more rigid.
- With XML, it is illegal to omit the closing tag. In HTML some elements do not have to have a closing tag to be able to present content in the way the user wants them to. For example:

```
<p>This is a paragraph
<p>This is another paragraph
```

In XML, however, all elements must have a closing tag, like this:

```
<p>This is a paragraph</p>
<p>This is another paragraph</p>
```

- Unlike HTML, XML tags are case sensitive.

With XML, the tag <Letter> is different from the tag <letter>.

Opening and closing tags must therefore be written with the same case:

```
<Message>This is incorrect</message>
<message>This is correct</message>
```

- Improper nesting of tags makes no sense to XML.

In HTML some elements can be improperly nested within each other and still display content in the desired way like this:

```
<b><i>This text is bold and italic</b></i>
```

In XML all elements must be properly nested within each other like this:

```
<b><i>This text is bold and italic</i></b>
```

- All XML documents must contain a single tag pair to define a root element.

All other elements must be within this root element. All elements can have sub elements (child elements). Sub elements must be correctly nested within their parent element:

```
<root>
<child>
<subchild>.....</subchild>
</child>
</root>
```

- With XML, it is illegal to omit quotation marks around attribute values.

XML elements can have attributes in name/value pairs just like in HTML. In XML the

attribute value must always be quoted. Study the two XML documents below. The first one is incorrect, the second is correct:

```
<?xml version="1.0" encoding="ISO-8859-1"?>
<note date=12/11/2002>
<to>Tove</to>
<from>Jani</from>
</note>
<?xml version="1.0" encoding="ISO-8859-1"?>
<note date="12/11/2002">
<to>Tove</to>
<from>Jani</from>
</note>
```

The error in the first document is that the date attribute in the note element is not quoted.

- With XML, the white space in your document is not truncated.

This is unlike HTML. With HTML, a sentence like this:

Hello my name is Tove,

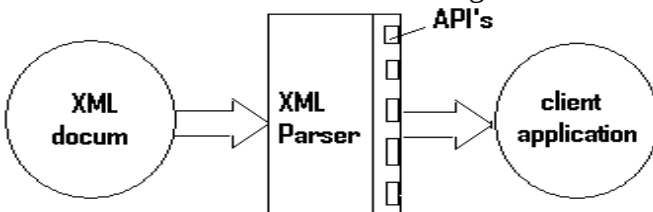
will be displayed like this:

Hello my name is Tove,

because HTML strips off the white space.

XML Parsing-

- It is a software library (or a package) that provides methods (or interfaces) for client applications to work with XML documents
- It checks the well-formattedness
- It may validate the documents
- It does a lot of other detailed things so that a client is shielded from that complexities



- DOM: Document Object Model
- SAX: Simple API for XML
- A DOM parser implements DOM API
- A SAX parser implements SAX API
- Most major parsers implement both DOM and SAX API's

The Document Object Model is an official recommendation of the World Wide Web Consortium (W3C). It defines an interface that enables programs to access and update the style, structure, and contents of XML documents. XML parsers that support the DOM implement that interface. When you parse an XML document with a DOM parser, you get back a tree structure that contains all of the elements of your document. The DOM provides a variety of functions you can use to examine the contents and structure of the document.

The DOM is a common interface for manipulating document structures. One of its design goals is that Java code written for one DOM-compliant parser should run on any other DOM-compliant parser without changes.

The DOM defines several Java interfaces. Here are the most common interfaces:

- **Node** - The base datatype of the DOM.
- **Element** - The vast majority of the objects you'll deal with are Elements.
- **Attr** Represents an attribute of an element.
- **Text** The actual content of an Element or Attr.
- **Document** Represents the entire XML document. A Document object is often referred to as a DOM tree.

SAX (the Simple API for XML) is an event-based parser for xml documents. Unlike a DOM parser, a SAX parser creates no parse tree. SAX is a streaming interface for XML, which means that applications using SAX receive event notifications about the XML document being processed an element, and attribute, at a time in sequential order starting at the top of the document, and ending with the closing of the ROOT element.

- Reads an XML document from top to bottom, recognizing the tokens that make up a well-formed XML document
- Tokens are processed in the same order that they appear in the document
- Reports the application program the nature of tokens that the parser has encountered as they occur
- The application program provides an "event" handler that must be registered with the parser
- As the tokens are identified, callback methods in the handler are invoked with the relevant information

Q5. Write short notes on:

a) AJAX

- AJAX stands for **A**synchronous **J**avaScript and **X**ML. AJAX is a new technique for creating better, faster, and more interactive web applications with the help of XML, HTML, CSS, and Java Script.
- Ajax uses XHTML for content, CSS for presentation, along with Document Object Model and JavaScript for dynamic content display.
- Conventional web applications transmit information to and from the sever using synchronous requests. It means you fill out a form, hit submit, and get directed to a new page with new information from the server.

- With AJAX, when you hit submit, JavaScript will make a request to the server, interpret the results, and update the current screen. In the purest sense, the user would never know that anything was even transmitted to the server.
- XML is commonly used as the format for receiving server data, although any format, including plain text, can be used.
- AJAX is a web browser technology independent of web server software.
- A user can continue to use the application while the client program requests information from the server in the background.
- Intuitive and natural user interaction. Clicking is not required, mouse movement is a sufficient event trigger.
- Data-driven as opposed to page-driven.

AJAX - Technologies

AJAX cannot work independently. It is used in combination with other technologies to create interactive webpages.

JavaScript

- Loosely typed scripting language.
- JavaScript function is called when an event occurs in a page.
- Glue for the whole AJAX operation.

DOM

- API for accessing and manipulating structured documents.
- Represents the structure of XML and HTML documents.

CSS

- Allows for a clear separation of the presentation style from the content and may be changed programmatically by JavaScript.

XMLHttpRequest

- JavaScript object that performs asynchronous interaction with the server.

Writing Browser Specific Code

The Simplest way to make your source code compatible with a browser is to use try...catch blocks in your JavaScript.

```
<html>
```

```
<body><script language="javascript" type="text/javascript">
```

```

<!--
//Browser Support Code
function ajaxFunction(){
    var ajaxRequest; // The variable that makes Ajax possible!

    try{
        // Opera 8.0+, Firefox, Safari
        ajaxRequest = new XMLHttpRequest();
    }catch (e){

        // Internet Explorer Browsers
        try{
            ajaxRequest = new ActiveXObject("Msxml2.XMLHTTP");
        }catch (e) {
            try{
                ajaxRequest = new ActiveXObject("Microsoft.XMLHTTP");
            }catch (e){

                // Something went wrong
                alert("Your browser broke!");
                return false;
            }
        }
    }
}
//-->
</script>
<form name='myForm'>
    Name: <input type='text' name='username' /> <br />
    Time: <input type='text' name='time' />
</form>
</body>

```

Steps of AJAX Operation

- A client event occurs.
- An XMLHttpRequest object is created.
- The XMLHttpRequest object is configured.
- The XMLHttpRequest object makes an asynchronous request to the Webserver.
- The Webserver returns the result containing XML document.
- The XMLHttpRequest object calls the callback() function and processes the result.
- The HTML DOM is updated.

AJAX - Issues

AJAX is growing very fast and that is the reason that it contains many issues with it. We hope with the passes of time, they will be resolved and AJAX will become ideal for web applications. We are listing down a few issues that AJAX currently suffers from.

Complexity is increased

- Server-side developers will need to understand that presentation logic will be required in the HTML client pages as well as in the server-side logic.
- Page developers must have JavaScript technology skills.

AJAX-based applications can be difficult to debug, test, and maintain

- JavaScript is hard to test - automatic testing is hard.
- Weak modularity in JavaScript.
- Lack of design patterns or best practice guidelines yet.

Toolkits/Frameworks are not mature yet

- Most of them are in beta phase.

No standardization of the XMLHttpRequest yet

- Future version of IE will address this.

No support of XMLHttpRequest in old browsers

- Iframe will help.

JavaScript technology dependency and incompatibility

- Must be enabled for applications to function.
- Still some browser incompatibilities exist.

JavaScript code is visible to a hacker

- Poorly designed JavaScript code can invite security problems.

b) **EJB**

Enterprise JavaBeans (EJB) is the server-side and platform-independent Java application programming interface (API) for Java Platform, Enterprise Edition (Java EE). EJB is used to simplify the development of large distributed applications.

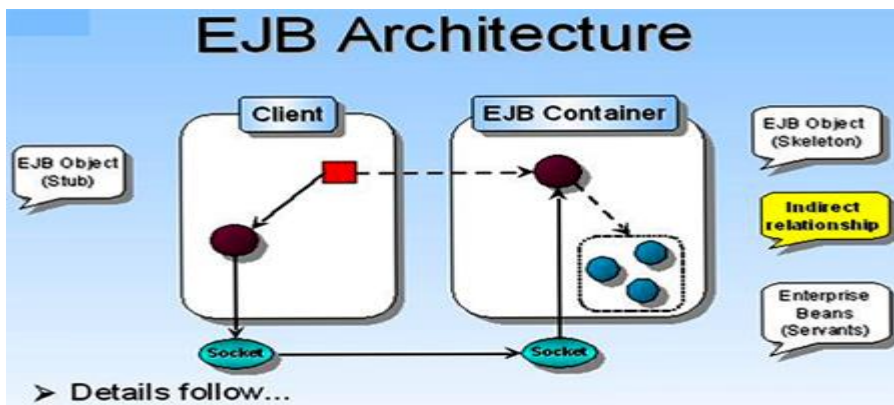
Enterprise JavaBeans (EJB) are reusable Java components that implement business logic and enable you to develop component-based distributed business applications.

EJBs reside in an EJB container, which provides a standard set of services such as

- Persistence,
- Security,
- Transactions, and
- Concurrency.

Enterprise JavaBeans are the standard for defining server-side components. WebLogic Server's implementation of the Enterprise JavaBeans component architecture is based on Sun Microsystems EJB specification.

Simple architecture of EJB



The EJB specification defines the following **four** types of Enterprise JavaBeans:

- **Stateless session.** These non-persistent EJBs provide a service without storing an interaction or conversation state between methods.
- **Stateful session.** These non-persistent EJBs maintain state across methods and transactions. Each instance is associated with a particular client.
- **Entity.** These persistent EJBs represent an object view of the data, usually rows in a database. An entity bean has a primary key as a unique identifier.
- **Message-driven.** These EJBs are integrated with the Java Message Service (JMS) to enable message-driven beans to act as a standard JMS message consumer and perform asynchronous processing between the server and the JMS message producer.

From the EJB sub-node, of the Deployments node of the Administration Console, you can update or configure EJBs deployed on WebLogic Server or monitor the performance.

EJB Architecture

There are three types of enterprise beans:

- Session beans
- Entity beans
- Message-driven beans.

Session beans: Session beans are non-persistent enterprise beans. They can be stateful or stateless.

- **A stateful session** bean acts on behalf of a single client and maintains client-specific session information (called conversational state) across multiple method calls and transactions. It exists for the duration of a single client/server session.
- **A stateless session** bean, by comparison, does not maintain any conversational state. Stateless session beans are pooled by their container to handle multiple requests from multiple clients.

Entity beans: Entity beans are enterprise beans that contain persistent data and that can be saved in various persistent data stores. Each entity bean carries its own identity.

- **Bean-managed persistence (BMP)** Entity beans that manage their own persistence are called bean-managed persistence (BMP) entity beans.
- **Container-managed persistence (CMP)** Entity beans that delegate their persistence to their EJB container are called container-managed persistence (CMP) entity beans.

Message-driven beans: Message-driven beans are enterprise beans that receive and process JMS

messages. Unlike session or entity beans, message-driven beans have no interfaces. They can be accessed only through messaging and they do not maintain any conversational state. Message-driven beans allow asynchronous communication between the queue and the listener, and provide separation between message processing and business logic.

Remote client view

The remote client view specification became available beginning with EJB 1.1. The remote client view of an enterprise bean is location independent. A client running in the same JVM as a bean instance uses the same API to access the bean as a client running in a different JVM on the same or different machine.

Remote interface: The remote interface specifies the remote business methods that a client can call on an enterprise bean.

Remote home interface: The remote home interface specifies the methods used by remote clients for locating, creating, and removing instances of enterprise bean classes.

Local client view

Unlike the remote client view, the local client view of a bean is location dependent. Local client view access to an enterprise bean requires both the local client and the enterprise bean that provides the local client view to be in the same JVM.

- The local client view therefore does not provide the location transparency provided by the remote client view.
- Local interfaces and local home interfaces provide support for lightweight access from enterprise bean that are local clients.
- Session and entity beans can be tightly couple with their clients, allowing access without the overhead typically associated with remote method calls.

The local client view specification is available in EJB 2.0 or later.

Local interface: The local interface is a lightweight version of the remote interface, but for local clients. It includes business logic methods that can be called by a local client.

EJB client JAR file

An EJB client JAR file is an optional JAR file that can contain the client interfaces that a client program needs to use the client views of the enterprise beans that are contained in the EJB JAR file. If you decide not to create an EJB client JAR file for an EJB module, all of the client interface classes will be in the EJB JAR file. By default, the workbench creates EJB client JAR projects for each corresponding EJB project.

EJB container

An EJB container is a runtime environment that manages one or more enterprise beans. The EJB container manages the life cycles of enterprise bean objects, coordinates distributed transactions, and implements object security. Generally, each EJB container is provided by an EJB server and contains a set of enterprise beans that run on the server.

Deployment descriptor

A deployment descriptor is an XML file packaged with the enterprise beans in an EJB JAR file or an EAR file. It contains metadata describing the contents and structure of the enterprise beans, and runtime transaction and security information for the EJB container.

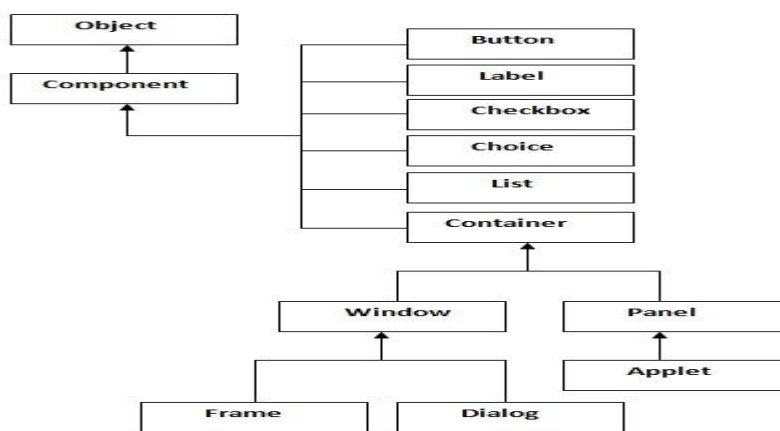
EJB server

An EJB server is a high-level process or application that provides a runtime environment to support the execution of server applications that use enterprise beans. An EJB server provides a JNDI-accessible naming service, manages and coordinates the allocation of resources to client applications, provides

access to system resources, and provides a transaction service. Although a message-driven EJB cannot be called directly by clients, it can call other EJBs itself. Message-driven EJBs are the only type that do not follow a strict request-response interaction with clients.

c) AWT

Java AWT (Abstract Windowing Toolkit) is an API to develop GUI or window-based application in java. Java AWT components are platform-dependent i.e. components are displayed according to the view of operating system. AWT is heavyweight i.e. its components uses the resources of system. The java.awt package provides classes for AWT api such as TextField, Label, TextArea, RadioButton, CheckBox, Choice, List etc. The hierarchy of Java AWT classes are given below.



Container

The Container is a component in AWT that can contain another components like buttons, textfields, labels etc. The classes that extends Container class are known as container such as Frame, Dialog and Panel.

Window

The window is the container that have no borders and menu bars. You must use frame, dialog or another window for creating a window.

Panel

The Panel is the container that doesn't contain title bar and menu bars. It can have other components like button, textfield etc.

Frame

The Frame is the container that contain title bar and can have menu bars. It can have other components like button, textfield etc.

Useful Methods of Component class

Method	Description
public void add(Component c)	inserts a component on this component.
public void setSize(int width,int height)	sets the size (width and height) of the component.
public void setLayout(LayoutManager m)	defines the layout manager for the component.

public void setVisible(boolean status)	changes the visibility of the component, by default false.
--	--

Java AWT Example

To create simple awt example, you need a frame. There are two ways to create a frame in AWT.

- By extending Frame class (inheritance)
- By creating the object of Frame class (association)

AWT Example by Inheritance

```
import java.awt.*;
class First extends Frame{
    First(){
        Button b=new Button("click me");
        b.setBounds(30,100,80,30);// setting button position
        add(b);//adding button into frame
        setSize(300,300);//frame size 300 width and 300 height
        setLayout(null);//no layout manager
        setVisible(true);//now frame will be visible, by default not visible
    }
    public static void main(String args[]){
        First f=new First();
    }
}
```

The `setBounds(int xaxis, int yaxis, int width, int height)` method is used in the above example that sets the position of the awt button.

AWT Example by Association

```
import java.awt.*;
class First2{
    First2(){
        Frame f=new Frame();
        Button b=new Button("click me");
        b.setBounds(30,50,80,30);
        f.add(b);
        f.setSize(300,300);
        f.setLayout(null);
        f.setVisible(true);
    }
    public static void main(String args[]){
        First2 f=new First2();
    }
}
```